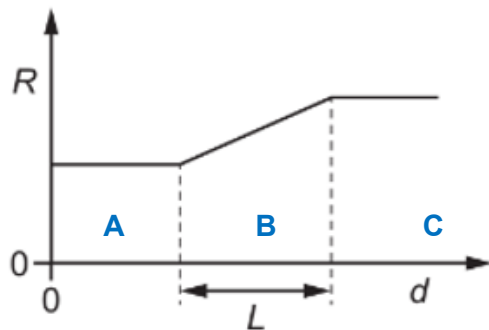
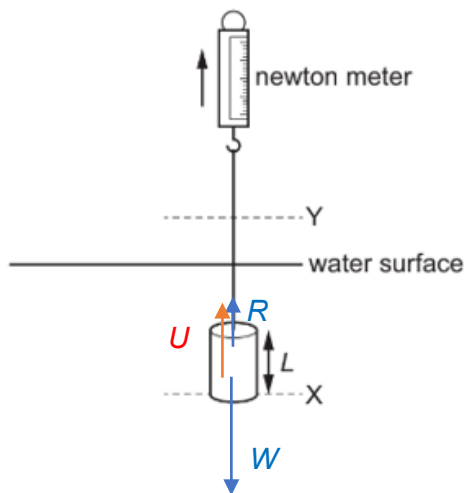


6

C



At the initial position, the forces acting on the cylinder are weight W , upthrust U and tension R . Magnitude of the sum of upward forces U and R is equal to downward weight W . As the metal cylinder is slowly raised,

In region A: cylinder is still fully submerged in water hence upthrust U remains the same, and weight W always remain the same throughout, hence tension R is constant.

In region B: cylinder is being pulled out of the water so upthrust U decreases at a constant rate due to the volume submerged in water decreases at a constant rate. Hence, R increases at a constant rate.

In region C: cylinder is fully out of the water hence upthrust U is zero so $W = R$ remains constant.

7

B

Option A: Wrong because only when the force is reduced and the rubber band follows the